Solution Assignment # 11

Modern Physics VI

Chapter 37: Relativity

Question 1 Solution:

The lifetime measured in the muon frame is the proper time that is $\Delta t_o \times$ u = 0.900c is the speed of the muon frame relative to the laboratory frame. The distance the particle travels in the lab frame is its speed in that frame times its lifetime in that frame.

(a) $\gamma = \frac{1}{\sqrt{1-0.9^2}} = 2.29$
So $\Delta t = \gamma\ \Delta t_o = 2.29 \times 2.29 \times 10^{-6}$ s $= 5.05 \times 10^{-6}$ s
(b) d = v $\Delta t = 0.900 \times 3.00 \times 10^{-6}$ s = 1.36 km

Question 2 Solution:

$l = l_o\sqrt{1-\left(\frac{u}{c}\right)^2}$

The length measured when the spacecraft is moving is l = 74 m, and l_o is the length measured in a frame at rest relative to the spacecraft.

$l_o = \frac{l}{\sqrt{1-\left(\frac{u}{c}\right)^2}} = \frac{74.0\ mm}{\sqrt{1-\left(\frac{0.600c}{c}\right)^2}} = 92.5$ m

$l_o > l$, the moving spacecraft appears to an observer on the planet to be shortened along the direction of motion.

Question 3 Solution:

There is a Doppler effect in the frequency of the radiation due to the motion of the star. And, as the star is moving away from the earth, so

$f = \sqrt{\frac{c-u}{c+u}}f_o = \sqrt{\frac{c-0.600c}{c+0.600c}}f_o = 0.500f_o = 0.500 \times 8.64 \times 10^{-6} = 4.32 \times 10^{14}$ Hz

The earth observer measures o lower frequency than the star emits because the star is moving away from the earth.

Question 4 Solution:

The rest energy is mc^2

(a) $K = \frac{mc^2}{\sqrt{1-\left(\frac{v}{c}\right)^2}} - mc^2 = mc^2$
$\frac{1}{\sqrt{1-\left(\frac{v}{c}\right)^2}} = 2$
By solving this equation we will get
v = 0.866c
(b) According to condition given in question
$K = 5mc^2 \Rightarrow \frac{1}{\sqrt{1-\left(\frac{v}{c}\right)^2}} = 6$
By solving this equation for v
v = 0.986c

so if v<<c, then K is much less than the rest energy of the particle.

Question 5 Solution:

(a) $E = mc^2 + K$

$E = 4.00mc^2$ it means that $K = 3.00\ mc^2 = 4.50 \times 10^{-10}$ J

(b) $E^2 = (mc^2)^2 + (pc)^2$

$E = 4.00\ mc^2$

$15.0\ (mc^2)^2 = (pc)^2$

$P = 1.98 \times 10^{-18}$ kg. m/s

(c) $E = mc^2\sqrt{1-\left(\frac{v}{c}\right)^2} = 0.968c$